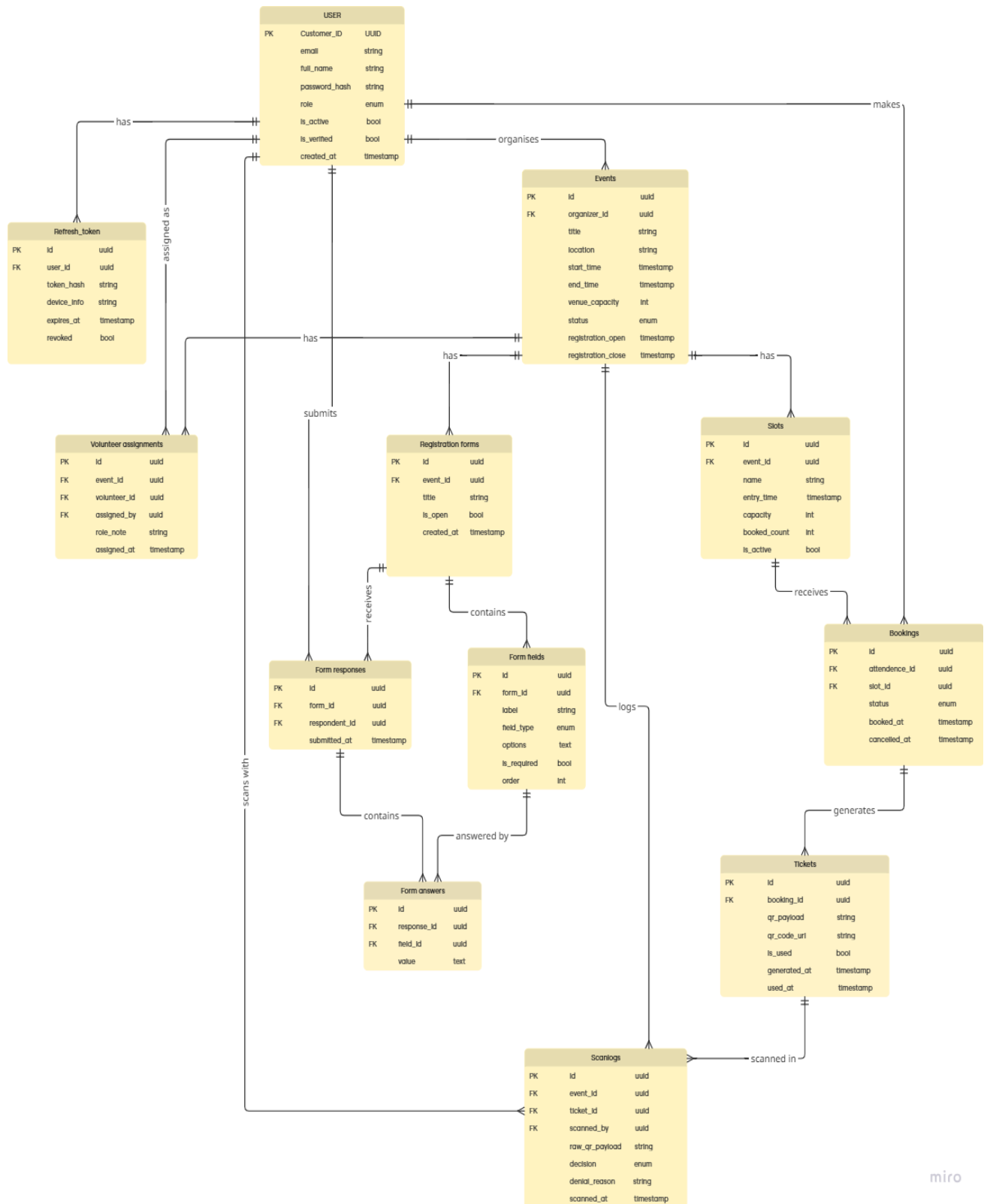


CROWD CONTROL-ER DIAGRAM



USERS

Central table for all roles — attendee, volunteer, organiser, admin

Column	Type	Key	Description
id	uuid	PK	Unique user identifier
email	string	UNIQUE	Login email, always lowercase
full_name	string		Display name
phone	string		Optional contact number
password_hash	string		bcrypt hash — never plaintext
role	enum		attendee / volunteer / organiser / admin
is_active	bool		False = soft ban, preserves history
is_verified	bool		Email verification status
created_at	timestamp		Account creation time
updated_at	timestamp		Last profile update

REFRESH TOKENS

Stores hashed refresh tokens per device, enabling per-device logout and token rotation

Column	Type	Key	Description
id	uuid	PK	Token record identifier
user_id	uuid	FK → users.id	Owner of the token
token_hash	string	UNIQUE	SHA-256 hash of the raw JWT — not reversible
device_info	string		User-agent string of the device
expires_at	timestamp		When this token becomes invalid
revoked	bool		True = manually invalidated (logout/password change)
created_at	timestamp		When token was issued

EVENTS

Events created by an organiser or admin, with booking windows and capacity limits

Column	Type	Key	Description
id	uuid	PK	Unique event identifier
organiser_id	uuid	FK → users.id	User who created the event
title	string		Event name

Column	Type	Key	Description
description	text		Full event description
location	string		Venue or address
start_time	timestamp		Event start (must be after end of registration)
end_time	timestamp		Event end (must be after start_time)
venue_capacity	int		Hard ceiling for total attendees
status	enum		draft / published / ongoing / completed / cancelled
registration_open	timestamp		When attendees can start booking
registration_close	timestamp		Booking deadline (must be $\leq$ start_time)
banner_image_url	string		URL to event banner image
created_at	timestamp		Record creation time
updated_at	timestamp		Last edit time

SLOTS

Entry time windows within an event (e.g. Gate A — 10:00 AM). Each slot has its own capacity.

Column	Type	Key	Description
id	uuid	PK	Unique slot identifier
event_id	uuid	FK → events.id	Parent event
name	string		Slot label e.g. "Gate A - 10:00 AM"
entry_time	timestamp		When attendees in this slot may enter
exit_time	timestamp		When this slot closes (optional)
capacity	int		Max attendees for this slot
booked_count	int		Live count of confirmed bookings (denormalised for fast reads)
is_active	bool		False = slot hidden from attendees
created_at	timestamp		Record creation time

BOOKINGS

One booking = one attendee reserving one slot. Unique per (attendee, slot).

Column	Type	Key	Description
id	uuid	PK	Unique booking identifier
attendee_id	uuid	FK → users.id	The attendee who booked
slot_id	uuid	FK → slots.id	The slot being reserved
status	enum		confirmed / cancelled / used
booked_at	timestamp		When the booking was made
cancelled_at	timestamp		Set when status changes to cancelled

Unique constraint: (attendee_id, slot_id) — prevents double booking the same slot

TICKETS

One ticket per confirmed booking. Holds the QR code used for entry scanning.

Column	Type	Key	Description
id	uuid	PK	Unique ticket identifier
booking_id	uuid	FK → bookings.id	Parent booking (one-to-one)
qr_payload	string	UNIQUE	Opaque UUID encoded in the QR — not guessable
qr_code_url	string		Path to QR image in file storage
is_used	bool		Flips to True on first valid scan
generated_at	timestamp		When the QR was generated
used_at	timestamp		When the ticket was scanned successfully

SCAN LOGS

Immutable audit log of every QR scan attempt. Never updated or deleted — insert only.

Column	Type	Key	Description
id	uuid	PK	Unique scan event identifier
event_id	uuid	FK → events.id	Which event the scan happened at
ticket_id	uuid	FK → tickets.id	Ticket that was scanned (null if QR was invalid)
scanned_by	uuid	FK → users.id	Volunteer or admin who scanned
raw_qr_payload	string		Exact string scanned — useful for logging invalid QRs

Column	Type	Key	Description
decision	enum		allowed / buffer / denied
denial_reason	string		Human-readable reason when decision ≠ allowed
scanned_at	timestamp		Exact time of scan
device_id	string		Scanner device identifier

VOLUNTEER ASSIGNMENTS

Links a volunteer to an event. Organiser assigns via POST /volunteers.

Column	Type	Key	Description
id	uuid	PK	Unique assignment identifier
event_id	uuid	FK → events.id	The event being staffed
volunteer_id	uuid	FK → users.id	The assigned volunteer
assigned_by	uuid	FK → users.id	Organiser who made the assignment
role_note	string		Optional note e.g. "Gate A scanner"
assigned_at	timestamp		When the assignment was made

Unique constraint: (event_id, volunteer_id) — a volunteer can't be assigned to the same event twice

REGISTRATION FORMS

Optional form per event, published on the website for attendees to fill before booking.

Column	Type	Key	Description
id	uuid	PK	Unique form identifier
event_id	uuid	FK → events.id	Parent event (one-to-one)
title	string		Form heading shown to attendees
is_open	bool		False = form closed, no new submissions
created_at	timestamp		Record creation time

Unique constraint: event_id — strictly one form per event

FORM FIELDS

Individual questions/fields inside a registration form. Ordered by order column.

Column	Type	Key	Description
id	uuid	PK	Unique field identifier
form_id	uuid	FK → registration_forms.id	Parent form
label	string		Question text shown to user
field_type	enum		text / number / choice / checkbox / date
options	text		JSON array of choices (only for choice type)
is_required	bool		Whether this field must be answered
order	int		Display order within the form

FORM RESPONSES

One submission of a registration form by an attendee. One per user per form.

Column	Type	Key	Description
id	uuid	PK	Unique response identifier
form_id	uuid	FK → registration_forms.id	The form that was submitted
respondent_id	uuid	FK → users.id	Attendee who submitted
submitted_at	timestamp		Submission time

Unique constraint: (form_id, respondent_id) — one response per user per form

FORM ANSWERS

Individual field answers inside a form response. One row per field per response.

Column	Type	Key	Description
id	uuid	PK	Unique answer identifier
response_id	uuid	FK → form_responses.id	Parent response
field_id	uuid	FK → form_fields.id	The field being answered
value	text		The answer value as a string

Unique constraint: (response_id, field_id) — one answer per field per response